POE Question Bank
Semester II 2024-2025

Part A

1. Quantity demanded for a good is given by the function

$$Q_A^d = 40 - 0.02P_A + 0.9P_B$$

   Where:

   - P_A = price of commodity A
   - P_B = price of commodity B

   What is the quantity demanded when the prices of good A and B are \$20 and \$8 respectively?

2. A consumer's Utility Function is given by

$$U(x, y) = x^2y$$

   What is the total utility obtained from a consumption bundle of (3,2)?

3. The price of a product increases from \$5 to \$7, and the quantity demanded decreases from 100 units to 80 units. Calculate the price elasticity of demand.

4. A producer is willing to sell a good for \$15 but manages to sell it at the market price of \$25. Calculate the producer surplus per unit.

5. Cost Function for a firm is given by

$$C = 5Q^3 + 8Q - ln(Q)$$

   What is the cost incurred by firm to produce one additional unit at a production level of 12 units?

6. A firm producing commodity A incurs fixed costs of \$40 and average variable costs of \$5. Market price of A is \$10. How many units must the firm sell to break even?

7. Demand and Supply Functions are given by

$$Q^d = 16 - 0.2P \qquad (1)$$
$$Q^s = 0.4P - 11 \qquad (2)$$

   Find the equilibrium price and quantity.

8. A firm produces in perfectly competitive market where the price is \$10. If the firm's cost function is given by

$$C = (Q + 3)^2$$

   Find the profit maximizing quantity.

9. The supply function of a firm is

$$Q^s = 2P - 10$$

   If the market clearing price is \$20, calculate the equilibrium quantity and producer surplus.

10. A firm produces two goods, X and Y. The cross-price elasticity of demand between them is -0.8. Comment on the nature of these goods. If the price of X increases by 10%, what is the percentage change in the quantity demanded of Y?

Part B

1. If the demand function for good x is given by

$$Q^d = 6 - 2P_x + 4P_y$$

where the prices of good y are fixed at \$1. Find the inverse demand function and consumer surplus if the price of good x is \$3?

2. Quantity demanded for a good is given by the function

$$Q_A^d = 40 - 0.02P_A + 0.9P_B + 0.3Y_d$$

Where:

- P_A = price of commodity A
- P_B = price of commodity B
- Y_d = disposable income

Find price elasticity of demand when the price of good A is \$20, price of good B is \$8 and consumer's disposable income is \$500.

3. A firm incurs fixed costs of \$300. Variable Cost function is given by

$$C_v = 50Q^2 + 13Q$$

and the Total Revenue function is given by

$$R = 10Q^3 - 4Q^2$$

The firm produces 6 units. Calculate firm's loss and average total cost. Should the firm continue to operate in short run?

4. The price elasticity of demand and price elasticity of supply for a good are -0.4 and 0.5 respectively at an equilibrium price of \$5 and quantity of 16 units. Find the demand and supply functions.

5. Utility obtained from consumptions of good X and Y can be expressed as

$$U(x, y) = xy$$

If the consumer has \$40, and the prices of X and Y are \$4 and \$2, respectively, what is their optimum consumption bundle?

6. Bob quits his computer programming job, where he was earning a salary of \$120,000 per year, to start his own business in a building he owns and was previously renting out for \$30,000 per year. He incurs the following expenses

- Salary paid to himself - \$55,000
- Other Expenses - \$25,000

Find the accounting cost and economic cost associated with his business.

7. A company using only Labor(L) and Capital(K) to manufacture its goods follows the Cobb-Douglas production function given by

$$Q = 6L^{\frac{1}{2}}K^{\frac{2}{3}}$$

The company projects the total cost of production to be \$4200. Per unit cost of labor and capital are \$20 and \$120 respectively. What combination of capital and labor should it use?

8. A firm's production function is given by

$$F(K, L) = AK^{\alpha}L^{\beta}$$

where K is capital, L is labor, and $A, \alpha, \beta > 0$. Derive the conditions for which this function exhibits increasing, decreasing and constant returns to scale.

9. A monopolist faces the following demand curve

$$P = 100 - 2Q$$

The monopolist's total cost function is given by

$$TC = 20 + 3Q^2$$

Find the profit maximizing price and quantity.

10. Demand and Supply Functions are given by

$$Q^d = 2 - \frac{1}{4}P \tag{1}$$
$$Q^s = \frac{1}{2}P - \frac{5}{2} \tag{2}$$

If government intervention results in a price floor at \$7, then calculate consumer surplus, producer surplus and deadweight loss.

Part C

1. Mike consumes two goods A and B. The utility he gains from consumption of these products is given by

$$U(A, B) = 4A^2B$$

His monthly income is \$120 and prices of A and B are \$5 and \$8 respectively. Suppose the price of good A decreases by \$2, then calculate the following.

   (i) Income Effect on A

   (ii) Substitution Effect on B

2. Suppose Bob Auto Inc. is the only firm producing solar powered cars. It's demand and cost curves are as follows

$$P = 30 - 3Q \tag{1}$$
$$TC = 3Q^2 + 5 \tag{2}$$

Find the socially optimal level of production and the social cost of monopoly.

3. In a perfectly competitive market, the demand and supply functions are

$$Q^d = 200 - 5P \tag{1}$$
$$Q^s = 3P - 60 \tag{2}$$

Suppose government imposes a \$10 per unit tax on the producers, then calculate

   (i) Tax burden on Consumers

   (ii) Tax burden on Producers

4. The demand curve for a manufacturing company is given by

$$Q^d = 100 - 5P$$

It has two plants and the marginal cost for each plant is given by

$$MC_1 = 10 + 5Q_1$$
$$MC_2 = 15 + 5Q_2$$

Determine the profit-maximizing output level of each plant and equilibrium price.

5. Suppose two goods X and Y are perfect complements of each other, i.e. they are consumed together in a fixed ratio. It is known that the ratio of consumption $x : y$ is 3 : 2. Find U(x,y) and plot the indifference curve U(x,y) = 10.

6. Assume Pooja Wines is the only supplier in the local market. They face the following demand curve

$$P = 180 - 2Q$$

Their total cost function is given by

$$TC = Q^2 + 20$$

If the market structure was initially perfectly competitive with $P = \$90$, calculate the change in producer surplus, consumer surplus and deadweight loss.

7. If a firm is given a subsidy of \$5/unit produced, how would this affect the good's price? Also determine the benefit distribution between consumers and producers. It is known that the price elasticity of demand (ϵ_d) and price elasticity of supply (ϵ_s) is -0.8 and 1.2 respectively.

8. A consumer's utility function is given by

$$U(x, y) = ln(x) + y$$

This type of a utility function is also called the quasi-linear utility function. For an arbitrary budget line $M = p_x x + p_y y$, find an expression for the optimal consumption bundle (x^*, y^*). Also plot and interpret the graph of M against x^*.

9. In a perfectly competitive market, the demand and supply functions are

$$Q^d = 200 - 5P \quad (1)$$
$$Q^s = 3P - 60 \quad (2)$$

Suppose government imposes a \$10 per unit tax on the producers, then calculate

(i) Deadweight Loss from Taxation
(ii) Tax Revenue

10. Consider the case for commodities X and Y which are perfect substitutes of each other, with a linear utility function given by

$$U(x, y) = 3x + 4y$$

The following information is known:

- Price of $X = \$2$
- Price of $Y = \$5$
- Income = \$10

Graphically interpret and find the equation of the indifference curve with maximum utility.

11. Suppose a consumer's preferences are represented by a utility function $U(x, y) = x + ln(y)$. The consumer has an income $I = 50$, and initial prices are $P_x = 1$ and $P_y = 1$. The price of x increases from $P_x = 1$ to $P_x = 1.5$. What type of a good is x?

(a) Giffen Good
(b) Normal Good
(c) Inferior Good
(d) None of the above

12. If a consumer's initial basket given by (x,y) is (5,7). Suppose the price of good x decreases and his new basket is (8,7). It is known that during the shift, his decomposition basket was (10,5). What type of a good is x?

(a) Giffen Good
(b) Normal Good
(c) Inferior Good
(d) None of the above

13. A consumer maximizes their utility $U(x, y) = x^{0.3}y^{0.7}$, with an income $I = 120$ and initial prices are $P_x = 6$ and $P_y = 4$. The price of x decreases from $P_x = 6$ to $P_x = 3$. What is x?

    (a) Giffen Good

    (b) Normal Good

    (c) Inferior Good

    (d) None of the above

14. Suppose Checks&Co. is the only screw manufacturer in the market. Their inverse demand function is given by $P = 20 - Q$ and their marginal cost is given by $MC = 2Q$. If the government imposes a \$4/unit tax on the monopolist, then calculate changes in

    (i) Consumer Surplus

    (ii) Producer Surplus

    (iii) Deadweight Loss

15. The price elasticity of demand and price elasticity of supply for a good are -0.4 and 0.5 respectively at an equilibrium price of \$5 and quantity of 16 units. Determine the new equilibrium price and quantity if demand curve shifts to the right by 10%.

16. Consider the utility function

    $$U(x, y) = \sqrt{x^2 + y^2}$$

    The price of good x is \$3, price of good y is \$4 and your monthly income is \$50.

    (i) Using the tangency condition, find the optimal consumption bundle.

    (ii) Explain whether the bundle above truly maximizes utility. If not, what is the actual optimal consumption bundle?

17. Given the utility function and budget constraint

    $$U(x_1, x_2) = x_1^{\alpha} x_2^{\beta}$$
    $$p_1 x_1 + p_2 x_2 = M$$

    Find an expression for $U(x_1^*, x_2^*)$ where (x_1^*, x_2^*) represents the optimal consumption bundle.

18. A firm's production function is given by

    $$Q = 4K^{\frac{2}{7}}L^{\frac{5}{7}}$$

    and the input prices are r=\$12 and w=\$5. Determine the cost-minimizing combination of K and L if the firm wants to produce 150 units of output.

19. Adidas has just launched a new sneakers collection in the market. Due to uncertainty in the market, prices could either be $P_1 = 450 - 0.58Q$ or $P_2 = 350 - 0.85Q$. If probability of P_1 is 0.45, determine the optimal output to maximize expected profit if their marginal cost is constant and equal to 80.

20. A non linear demand function is given by $Q = aP^{-b}$, where a and b are positive constants. How does the point price elasticity of demand vary along the demand curve? If $\Delta P = -10\%$, find the percentage change in quantity demanded.